

Cambridge (CIE) IGCSE Physics
Paper 1: Multiple Choice (Set A)
Core

Saturday 26 April 2025

Afternoon (Time: 0 hours 45 minutes)

Total marks

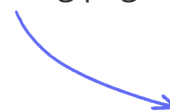
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Instructions

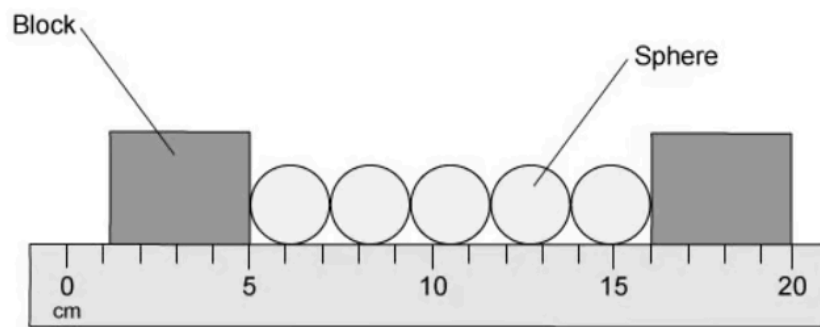
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 Five identical spheres are placed between two blocks in order to measure their diameter.

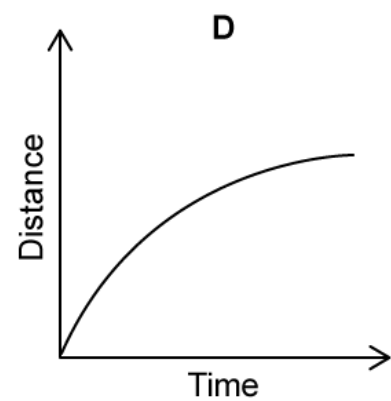
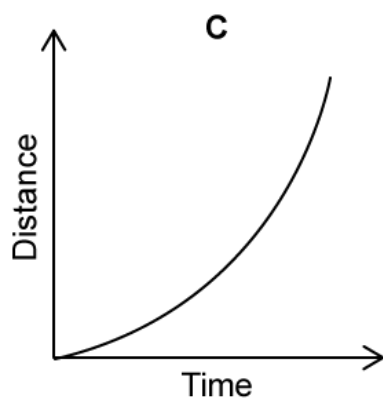
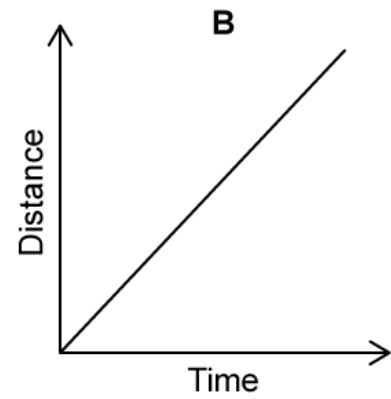
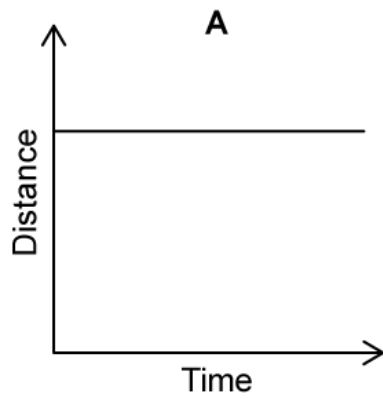


What is the diameter of a single sphere?

- A. 2.2 cm
- B. 2.0 cm
- C. 2.1 cm
- D. 2.3 cm

(1 mark)

2 Which distance-time graph shows an object moving with constant speed?



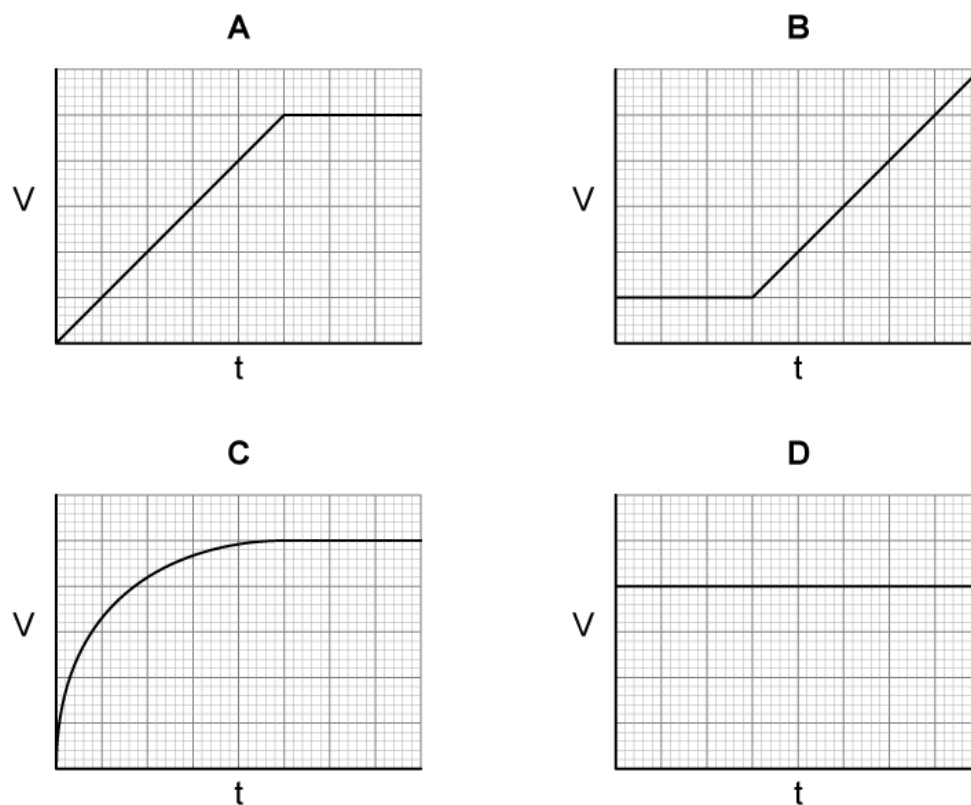
(1 mark)

3 Which row in the table gives the correct units for mass and for weight?

	Mass	Weight
A	kg	kg
B	N	kg
C	N	N
D	kg	N

(1 mark)

- 4 Four speed-time graphs have axes with the same scales. Each large square has a height of 1 m/s and width of 1 s.



Which graph shows the greatest distance covered?

(1 mark)

- 5 Some unknown liquid in a beaker has a mass of 200 g and a volume of 230 cm^3 .

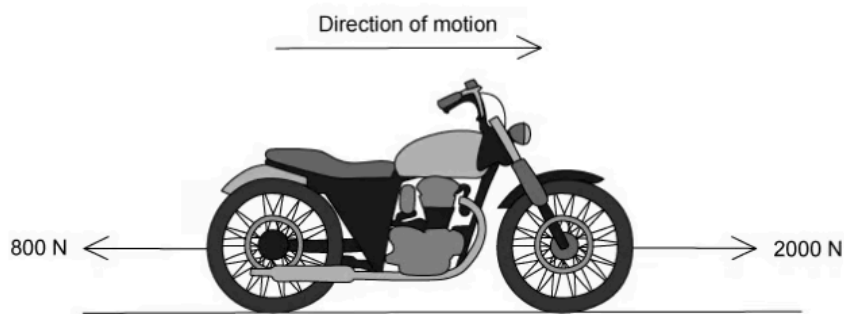
The density of water is 1.0 g/cm^3

How does the density of the unknown liquid compare with the density of water?

- A. Its density is greater than the density of water
- B. Its density is less than the density of water
- C. Its density is the same as the density of water
- D. With the information given, it is impossible to tell.

(1 mark)

- 6 A motorbike travels at a constant speed along a flat road. Three horizontal forces act on the motorbike. Two of them are shown in the diagram below.



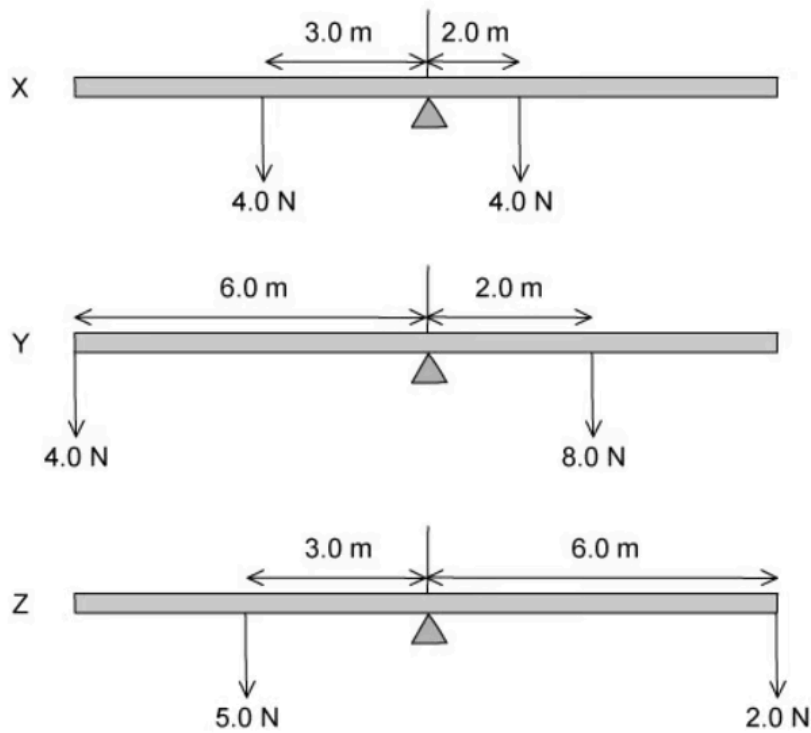
What is the size and direction of the third force?

- A. 2800 N forwards
- B. 1200 N backwards
- C. 1200 N forwards
- D. 2800 N backwards

(1 mark)

- 7 The diagrams below show three beams, each is pivoted at the centre. Each beam also

has two forces acting on it, as shown.

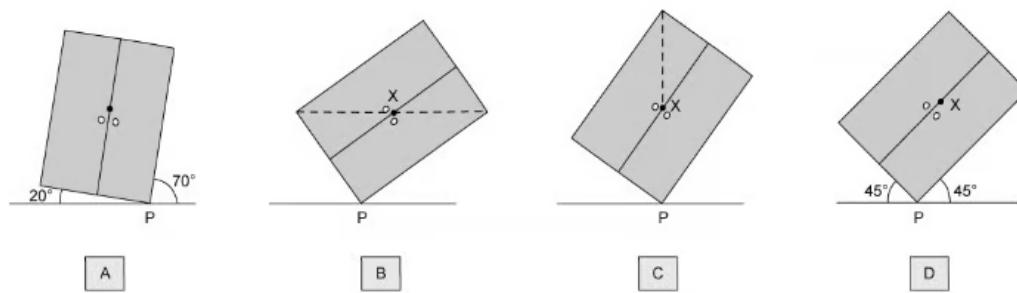


Which of the beams will rotate clockwise?

- A.** X, Y and Z
- B.** X and Y
- C.** Y and Z
- D.** None

(1 mark)

8 A wardrobe is shown tilted at a variety of angles.



In which position is the wardrobe in equilibrium?

(1 mark)

9 Which energy source relies on radiation from the Sun?

- A. Fossil fuels
- B. Nuclear fission
- C. Tidal power
- D. Geothermal energy

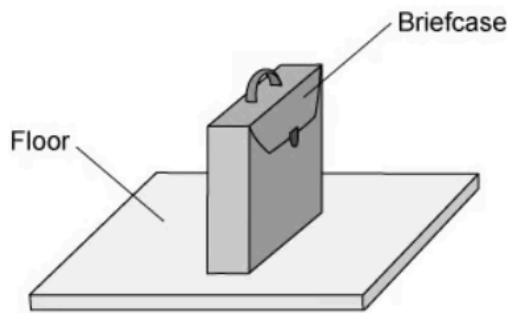
(1 mark)

10 Which of the following energy transfers is most efficient?

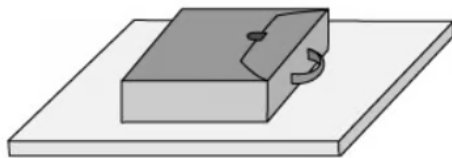
- A. Useful energy out = 70%
- B. Wasted energy out = 45%
- C. Useful energy out = 35%
- D. Wasted energy out = 25%

(1 mark)

- 11 A briefcase, with flat, rectangular sides rests on the floor as shown in the diagram.



The briefcase is now turned so that it rests with its large, flat side on the floor.



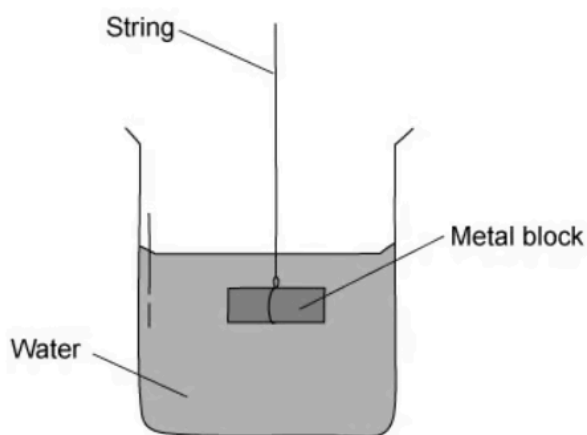
How has the change affected the force on the floor, and the pressure exerted by the briefcase on the floor?

	Force	Pressure
A	unchanged	decreased
B	unchanged	unchanged
C	decreased	decreased
D	decreased	unchanged

(1 mark)

- 12** A small metal block is suspended under the surface of a beaker of water by a string.

The metal block experiences a pressure exerted by the liquid.



What would increase the pressure exerted on the metal block?

- A.** increasing the surface area of the stone
- B.** using a liquid with a lower density
- C.** increasing the mass of the metal block
- D.** lowering the metal block deeper into the liquid

(1 mark)

- 13** A popular demonstration of kinetic theory is to view smoke particles through a microscope, while illuminating them with a bright light.

The diagram shows the view of the smoke particles through a microscope. They are moving about randomly.

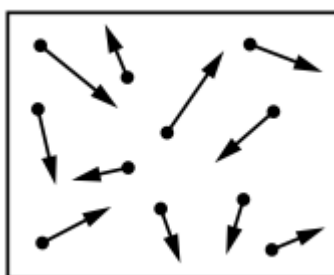


What causes the motion of the smoke particles?

- A.** The smoke particles are attracted to each other.
- B.** The smoke particles vibrate because they are warm.
- C.** Smaller air particles are colliding with the smoke particles.
- D.** The microscope is vibrating slightly, making the smoke particles *appear* to move.

(1 mark)

14 The diagram represents particles of a gas inside a closed container of constant volume.



The gas is heated.

What happens to the particles of the gas?

- A.** They expand.
- B.** They get closer together.
- C.** They hit the container walls with less force.
- D.** They move faster

(1 mark)

15 The water in outdoor swimming pools gradually evaporates.

How does the process of evaporation affect the temperature of the water in the swimming pool and what is the effect on the average kinetic energy of the water molecules?

	Temperature	Average kinetic energy per molecule
A	reduces	reduces
B	stays the same	stays the same
C	increases	increases
D	reduces	stays the same

(1 mark)

16 A car tyre is inflated with air.

As the car drives, the temperature of the tyre increases, which also causes the temperature of the air inside the tyre to increase. This increases the pressure of the air inside the tyre.

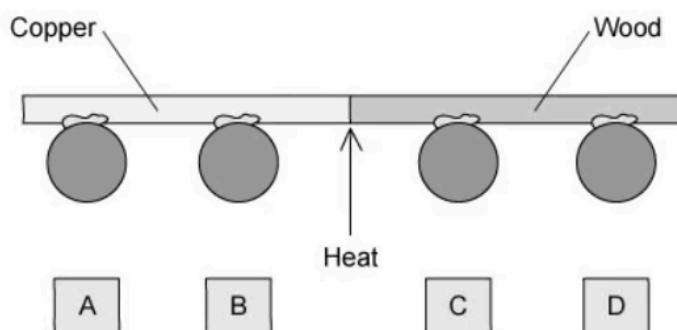
Which of the following is a correct explanation as to why?

- A.** The mass of air inside the tyre has increased.
- B.** The area of the tyre has decreased, thus increasing the pressure.
- C.** The air molecules collide with each other less frequently.
- D.** The average kinetic energy of the air molecules has increased.

(1 mark)

17 In a very common classroom Physics demonstration, a teacher sets up a rod, as shown in the diagram. Half of the rod is copper, the other half is wood.

She sticks four lead balls to the rod using wax. The wax melts when its temperature rises a small amount, causing the ball to fall off.

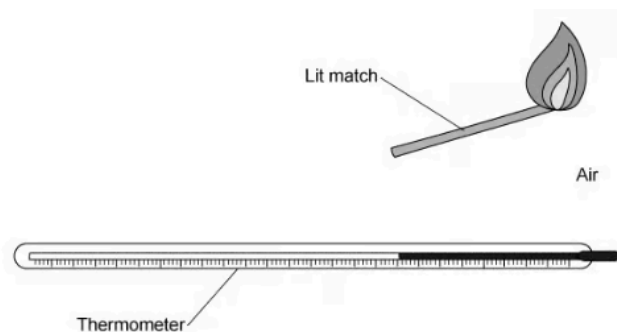


The teacher heats the rod in the centre, with something more gentle than a Bunsen burner, that will not burn the wood.

Which ball falls off first?

(1 mark)

18 A lit match is placed above a thermometer.



Which row in the table shows how the thermal energy from the lit match reaches the thermometer?

	Convection	Conduction	Radiation
A	yes	yes	yes
B	yes	yes	no
C	no	yes	yes
D	no	no	yes

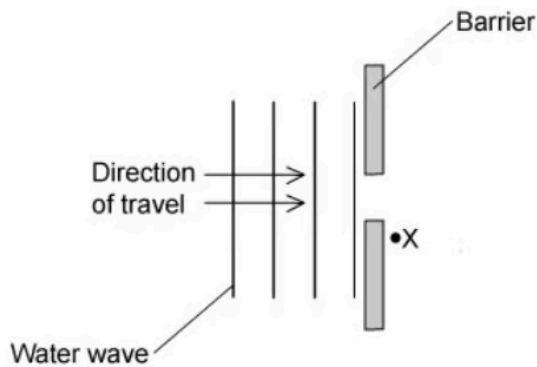
(1 mark)

19 What is the definition of the word **amplitude**?

- A. The number of complete waves that pass a point per second.
- B. The distance from the peak of one wave to the peak of an adjacent wave.
- C. The distance from the equilibrium position of a peak plus the same distance from a trough.
- D. The maximum distance from the equilibrium position of any point on a wave.

(1 mark)

- 20** What is the name given to the effect that allows the wavefronts to reach point X, shown on the diagram?



- A.** Reflection
- B.** Refraction
- C.** Diffraction
- D.** Dispersion

(1 mark)

- 21** A teacher has a clock mounted in such a way that students can only see its reflection in a plane mirror.

The diagram shows the reflection of the clock.



What time is shown on the clock?

- A.** 05:40

B. 07:40

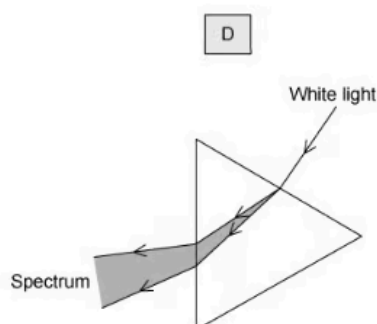
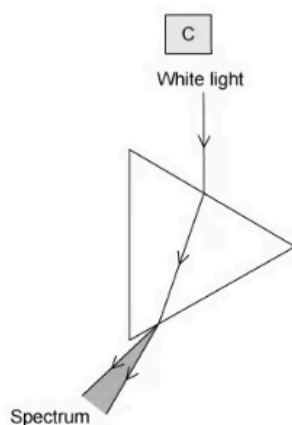
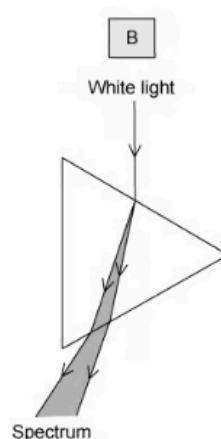
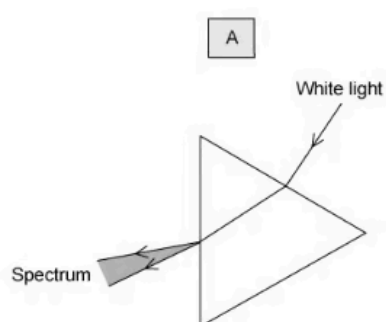
C. 07:20

D. 06:20

(1 mark)

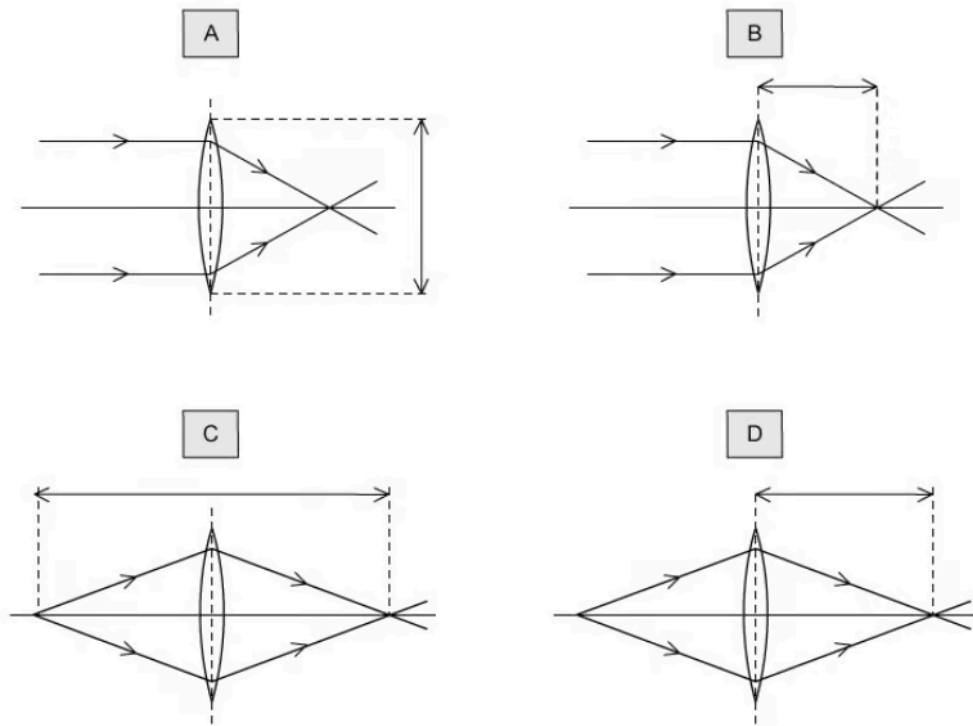
- 22 A physics teacher demonstrates dispersion for her class. She asks the students to draw a diagram of the dispersion.

The diagrams show four examples of what different students drew. One of them is correct. Which one?



(1 mark)

23



Which of the images correctly shows the focal length of the thin converging lens?

(1 mark)

24 Which of the following types of electromagnetic radiation has the highest speed in a vacuum?

- A. Microwaves
- B. Radio waves
- C. Visible light
- D. None of the above

(1 mark)

25 The speed of a wave is doubled. The frequency remains the same.

What happens to the wavelength of the wave?

- A.** It doubles
- B.** It halves
- C.** It remains the same
- D.** It becomes four times as large

(1 mark)

26 Which of the lists of metals below contains only metals which are non-ferrous?

- A.** Iron, steel, nickel
- B.** Cobalt, copper, steel
- C.** Aluminium, brass, copper
- D.** Iron, tin, aluminium

(1 mark)

27 Which statement about a permanent bar magnet is **incorrect**?

- A.** The S-pole attracts the N-pole of another magnet
- B.** The S-pole repels the S-pole of another magnet
- C.** It cannot repel a non-magnetic material
- D.** It is made from a soft magnetic material (such as soft iron)

(1 mark)

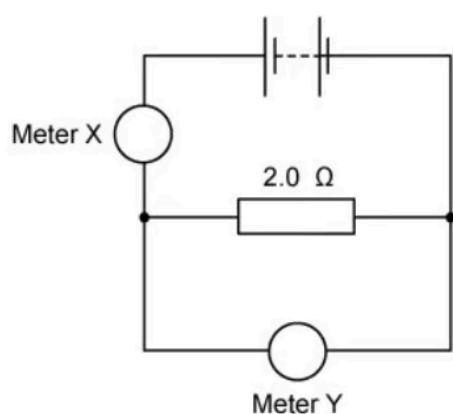
28 A PVC (plastic) rod is rubbed with a nylon cloth. This process causes electrons to be transferred between the rod and the cloth, causing both objects to become charged.

Which row gives the correct nature of the charges on both the cloth and the rod, and the effect the objects have on each other after becoming charged?

	Charges on rod and cloth	Effect
A	the same	repel
B	the same	attract
C	opposite	repel
D	opposite	attract

(1 mark)

29 A $2.0\ \Omega$ resistor is connected to meter X and meter Y as shown in the diagram.



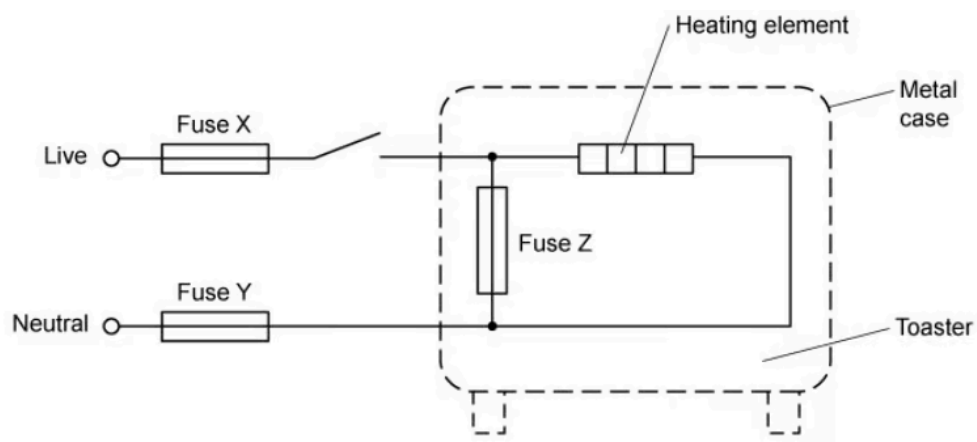
Which row in the table shows possible values for the two meters?

	X	Y
A	4.0	8.0
B	2.0	2.0
C	4.0	2.0
D	1.0	1.0

(1 mark)

30 The diagram shows a toaster that has been wired up.

Three fuses have been connected to the toaster, but only one or two have been connected correctly.

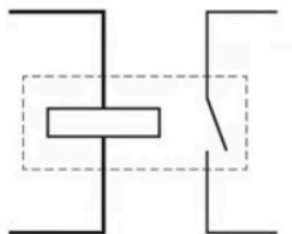


Which of the fuses has been positioned correctly?

- A.** Fuse X and fuse Y
- B.** Fuse X only
- C.** Fuse Z only
- D.** Fuse X and fuse Z

(1 mark)

31 This is the circuit symbol for which component?



- A. A fuse
- B. A buzzer
- C. A transistor
- D. A relay

(1 mark)

32 An oven is wired into somebody's kitchen. The oven is protected by a fuse, and functions normally, but the owner notices that the wires that connect the oven to the mains electricity supply get quite hot.

What could be done to reduce the temperature of the wires?

- A. Add a circuit breaker to the circuit.
- B. Put thicker insulation on the wires.
- C. Replace the wires with thicker diameter wires.
- D. Replace the wires with thinner diameter wires.

(1 mark)

33 When a wire is moved through a magnetic field, an e.m.f. is induced in the wire.

Which of the following devices uses this effect?

- A.** electric motor
- B.** electricity generator
- C.** photon accelerator
- D.** transformer

(1 mark)

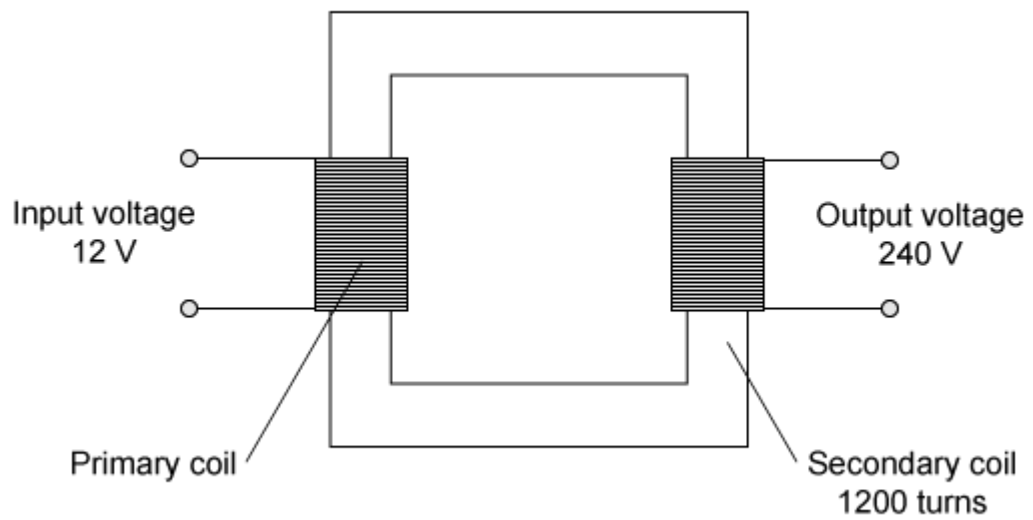
34 After electricity is generated at a power station, the voltage is increased significantly for transmission across the country via the national grid.

What is the advantage of transmitting electricity at very high voltages?

- A.** It makes the electricity flow more quickly.
- B.** It increases the efficiency of the electricity transfer.
- C.** It produces more power.
- D.** It is safer to transmit electricity at high voltage.

(1 mark)

- 35** A transformer has 1200 turns on its secondary coil and has an output voltage of 240 V.



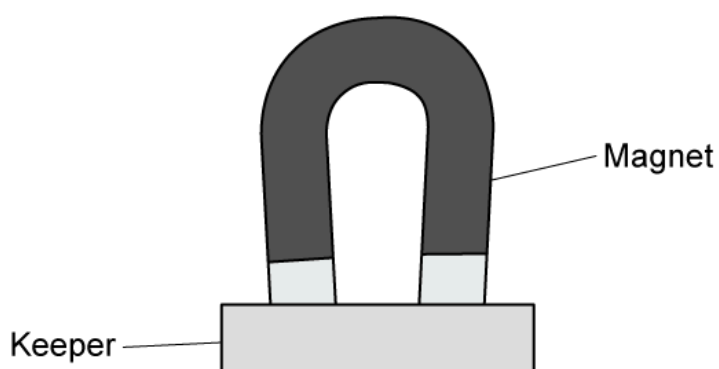
If the input voltage is 12 V, how many turns are there on the primary coil?

- A.** 2
- B.** 60
- C.** 1200
- D.** 24 000

(1 mark)

- 36** The diagram shows two bar magnets, stored with metal keepers across the ends. The

keepers help to keep the magnets magnetised.



The material used for the keepers becomes strongly magnetised when placed in contact with the magnets, but does not remain magnetised when taken away from the magnets.

What is a suitable metal to use for the magnets and what is a suitable metal to use for the keepers?

	Metal for keeper	Metal for magnet
A	steel	steel
B	iron	steel
C	steel	iron
D	iron	iron

(1 mark)

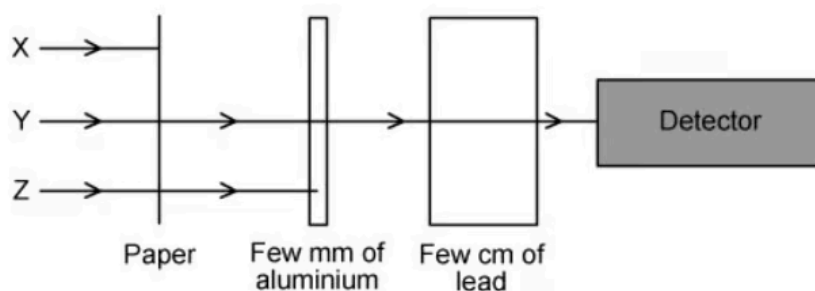
- 37** An atom consists of electrons orbiting a nucleus. The nucleus contains protons and neutrons.

For the atom to be neutral, which statement about the atom **must** be correct?

- A.** The number of electrons is equal to the number of protons.
- B.** The number of protons is equal to the number of neutrons.
- C.** The number of electrons, protons and neutrons are all different.
- D.** The number of electrons is equal to the number of neutrons.

(1 mark)

- 38** A student has three radioactive sources **X**, **Y** and **Z**. They devised the following experiment to determine what type of radiation each source is emitting.



The student found that the radiation from **X** was stopped by a sheet of paper, **Y** was partially stopped by a few cm of lead and **Z** was stopped by a few mm of aluminium.

Which of the following correctly identifies each type of radiation

	X	Y	Z
A	α -particle	β -particle	γ -ray
B	α -particle	γ -ray	β -particle
C	γ -ray	β -particle	α -particle
D	β -particle	γ -ray	α -particle

(1 mark)

39 What creates the periodic nature of the seasons?

- A.** The tilting of the Earth's axis
- B.** The orbiting of the Earth around the Sun
- C.** The combination of the orbiting of the Earth around the Sun and the Earth's tilting axis
- D.** The change in temperature of the Sun

(1 mark)

40 What is the correct description of the Milky Way?

- A.** Universe
- B.** Star
- C.** Galaxy
- D.** Nebula

(1 mark)